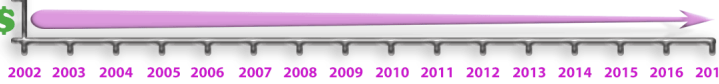




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Kinase Activities of RIPK1 and RIPK3 Can Direct IFN- β Synthesis Induced by Lipopolysaccharide

Danish Saleh,^{*,†} Malek Najjar,[‡] Matija Zelic,[§] Saumil Shah,[¶] Shoko Nogusa,^{||} Apostolos Polykratis,^{*,**,*††} Michelle K. Paczosa,^{‡‡} Peter J. Gough,^{§§} John Bertin,^{§§} Michael Whalen,^{¶¶} Katherine A. Fitzgerald,^{|||} Nikolai Slavov,^{##} Manolis Pasparakis,^{*,**,*††} Siddharth Balachandran,^{||} Michelle Kelliher,[§] Joan Mecsas,^{***} and Alexei Degterev^{*,†,‡,¶,||}

The innate immune response is a central element of the initial defense against bacterial and viral pathogens. Macrophages are key innate immune cells that upon encountering pathogen-associated molecular patterns respond by producing cytokines, including IFN- β . In this study, we identify a novel role for RIPK1 and RIPK3, a pair of homologous serine/threonine kinases previously implicated in the regulation of necroptosis and pathologic tissue injury, in directing IFN- β production in macrophages. Using genetic and pharmacologic tools, we show that catalytic activity of RIPK1 directs IFN- β synthesis induced by LPS in mice. Additionally, we report that RIPK1 kinase-dependent IFN- β production may be elicited in an analogous fashion using LPS in bone marrow-derived macrophages upon inhibition of caspases. Notably, this regulation requires kinase activities of both RIPK1 and RIPK3, but not the necroptosis effector protein, MLKL. Mechanistically, we provide evidence that necrosome-like RIPK1 and RIPK3 aggregates facilitate canonical TRIF-dependent IFN- β production downstream of the LPS receptor TLR4. Intriguingly, we also show that RIPK1 and RIPK3 kinase-dependent synthesis of IFN- β is markedly induced by avirulent strains of Gram-negative bacteria, *Yersinia* and *Klebsiella*, and less so by their wild-type counterparts. Overall, these observations identify unexpected roles for RIPK1 and RIPK3 kinases in the production of IFN- β during the host inflammatory responses to bacterial infection and suggest that the axis in which these kinases operate may represent a target for bacterial virulence factors. *The Journal of Immunology*, 2017, 198: 000–000.

Homologous serine/threonine kinases RIPK1 and RIPK3 are critical regulators of necroptosis, a form of necrotic cell death associated with pathologic tissue injury (1–6). RIPK1 and RIPK3 are components of the signaling machinery downstream of innate immune pathogen recognition receptors, TLR3 and TLR4 (7–11). Catalytic activity of these kinases, induced by activation of TLR3 or TLR4 in the presence of pan-caspase inhibitor, zVAD.fmk (zVAD), promote necroptosis in primary macrophages (9, 12–14). This regulation is dependent on the TLR3/4 Rip-homotypic interacting motif domain-containing adapter protein,

TRIF (9, 12–14). Upon activation, RIPK1 and RIPK3 form insoluble cellular aggregates, termed “necrosomes,” which function to allow phosphorylation and activation of the target pseudokinase, MLKL, by RIPK3 (15–18). Phosphorylation of MLKL promotes its oligomerization and translocation to the cell surface, where it functions to increase membrane permeability and facilitate necrotic cell death through a mechanism that is not yet fully understood (19–23).

New evidence suggests existence of direct links of RIPK1 and RIPK3 to inflammation that occur independently of cell death. For

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D.S. performed cell viability analyses, Western analyses, and ELISAs; M.N., D.S., and S.S. performed RNA work and qRT-PCRs; D.S., M.N., and M.Z. performed in vivo experiments; D.S. and N.S. fractionated lysates using sucrose gradient and velocity sedimentation; D.S. prepared the fractionation samples; M.K.P. generated and provided $\Delta cpsB$ bacterial strain; J.B. and P.J.G. generated and provided the K45A RIPK1 and K51A RIPK3 mice; A.P. and M.P. developed RIPK1 D138N mice; M.Z., M.P., and M.K. provided the D138N RIPK1 cells; M.W. provided *Ripk3*^{−/−} cells; S.N. and S.B. provided *Mkl1*^{−/−} cells; K.A.F. provided *Irf3*^{−/−}/*Irf7*^{−/−} cells; J.M. and M.P. provided bacterial strains; M.Z., J.B., P.J.G., M.P., K.A.F., N.S., S.B., and J.M. helped revise the manuscript; D.S. and A.D. wrote and revised the manuscript.

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Abbreviations used in this article: BMDM, bone marrow-derived macrophage; IFN-I, type I IFN; IRF, IFN regulatory factor; MOI, multiplicity of infection; NP-40, Nonidet P-40; PSA, penicillin/streptomycin/antimycotic; qRT-PCR, quantitative real-time PCR; WT, wild-type.

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example, in vitro, RIPK1 kinase activation has been linked to TNF- α production in contexts of caspase inhibition and DNA damage (24–27). We have recently demonstrated the existence of a TRIF-mediated cell death-independent signaling pathway downstream of RIPK1 and RIPK3 that involves activation of Erk1/2 and NF- κ B, and directs synthesis of acute inflammatory cytokines, including TNF- α , after TLR4 activation (14). In another example, RIPK1 kinase-dependent IL-1 α expression and tissue inflammation, occurring in the absence of necroptosis, were reported in the Ptpn (spin) model of human neutrophilic dermatitis (28). In myeloid-derived cells, RIPK3 is required for activation of NF- κ B and IL-1 β release in both a kinase-independent and -dependent manner conditional on caspase-8 inhibition (29–31).

In this article, we report a novel role for the kinase activity of RIPK1 in directing LPS-induced IFN- β synthesis in mice and in vitro in primary mouse macrophages when caspases are inhibited. We report that this regulation requires TRIF and occurs by way of a canonical type I IFN (IFN-I) response pathway, requiring TBK1, IKK ϵ , and IFN regulatory factor (IRF) 3/7. These canonical IFN-I-inducing factors colocalize to RIPK1/RIPK3 kinase containing detergent-insoluble cellular fractions after TLR4 activation, and this localization was disrupted by the selective RIPK1 kinase inhibitor, Nec-1s. Together, these data demonstrate that a necrosome-like RIPK1 and RIPK3 signaling platform mediates TRIF-dependent IFN- β production. Moreover, enhanced RIPK1 and RIPK3 kinase-dependent IFN- β synthesis was observed after challenge with attenuated mutant strains of *Yersinia pseudotuberculosis* and *Klebsiella pneumoniae* in the presence of caspase inhibition, suggesting that this pathway may represent a new target of bacterial virulence factors.

Materials and Methods

Animals

Female BALB/c mice at 6–8 wk of age were used for LPS experiments (Charles River Labs). *Ripk3*^{−/−} (on C57BL/6 background) and matched controls were previously described (32) and provided to us by Dr. V. Dixit (Genentech). *Sting*^{−/−} mice and matched controls were a gift from Dr. A. Poltorak. *TRIF*^{−/−} (*Ticam1*^{−/−}) (C57BL/6J-*Ticam1*Lps2/J) mice and corresponding control mice (C57BL/6J, B6.129P2) were purchased from Jackson Laboratories. *D138N RIPK1*, *K45A RIPK1*, *K51A RIPK3*, and *Mik1*^{−/−} mice were previously described (30, 33–35).

All use of animals was approved by the Tufts University, the University of Massachusetts, and Fox Chase Cancer Center Institutional Animal Care and Use Committees. Mice were maintained in Tufts animal facility in cages with light-dark cycle, and experiments were performed according to the protocol with all efforts to minimize the number and suffering of the animals.

Bacteria

Yersinia strains (*Y. pseudotuberculosis* and Δ yscF) were on the IP2666 background and have been previously described (36). *Yersinia* bacteria were cultured overnight in 2XYT at 26°C. On the day of infection, cultures were diluted 1:40 in 5 mM CaCl₂ containing 2XYT for 2 h and then moved to 37°C for 2 h before infection. *Klebsiella* strains (*K. pneumoniae* and Δ cpsB) were on the *K. pneumoniae* ATCC 43816 background. Δ cpsB was generated by M.K.P. and will be reported in a future publication. *Klebsiella* bacteria were grown overnight at 37°C in L broth.

Infections

Macrophages were infected in antibiotic-free media (10% L929 conditioned media + 10% FBS in RPMI 1640) bacteria at a multiplicity of infection (MOI) of 40–60. Following infection, bacteria and cells were spun down at 300 $\times$ g for 3 min. Two hours postinfection, 100 μ g/ml gentamicin was administered and macrophages were harvested 6 h postinfection. Before infection, bacterial counts were estimated using an OD readout and confirmed by CFU assay.

In vivo LPS challenge and Nec-1s treatment

Mice were injected i.v. or i.p. with 30 mg/kg optimized Nec-1s (7-Cl-O-Nec-1) (37), 15 min before i.p. injection of 50 μ g/kg LPS (Sigma). CD11b⁺ bone marrow cells were collected 1 h postinjection from femurs and stained

at a dilution of 1:300 using PE-conjugated anti-mouse/human CD11b⁺ (101207; BioLegend) for 1 h in FACS buffer (2% FBS and 1 μ M EDTA in PBS). Twenty to forty percent of total cells were CD11b⁺, and these were sorted by FACS. Sorted cells were used for quantitative real-time PCR (qRT-PCR) or RNA sequencing.

Reagents

LPS (*Escherichia coli* 0111:B4) was purchased from Sigma. Optimized Necrostatin-1s (Nec-1s) (5-[(7-chloro-1H-indol-3-yl) methyl]-3-methyl-2,4-imidazolidinedione) was synthesized as previously described (38). For in vivo administration, Nec-1s was dissolved in PBS containing 25% Polyethylene Glycol 400 (Spectrum Laboratories). RIPK3 inhibitor, GSK872, was also previously described (9, 39). TBK1/IKK ϵ inhibitors, MRT67307 and BX795, were previously described (40, 41). IDN6556 was previously described (42). zVAD.fmk was purchased from ApexBio.

Cells

Bone marrow-derived macrophages (BMDMs) were prepared by flushing bone marrows from femurs and tibias. BMDMs were allowed to differentiate for 7 d in the presence of conditioned media from L929 cells, containing M-CSF: 30% L929 media, 20% FBS, and 1% PSA (penicillin/streptomycin/antimycotic solution) in RPMI 1640. Medium was replenished on day 3. Twenty-four to forty-eight hours before treating the cells, the medium was exchanged to 10% L929 media, 10% FBS, and 1% PSA in RPMI 1640. On day 7, the adherent cells were collected using PBS and centrifuged at 430 $\times$ g for 5 min and replated for the experiments. For mRNA and Western blot experiments, 2 $\times$ 10⁶ BMDMs were seeded in six-well plates. For cell viability experiments, 50,000 cells per well were seeded in 96-well plates. RAW cells (RAW264.7) were grown in DMEM containing 10% FBS and 1% PSA. For experiments, cells were treated with all reagents (stimulants and inhibitors) simultaneously. For example, inhibitors (i.e., Nec-1s) were added to cells with LPS and zVAD or IDN at time zero.

Western blotting

Cells were lysed and harvested in RIPA buffer (Cell Signaling) supplemented with PMSF (5 μ g/ml; Sigma), leupeptin (1 μ g/ml), pepstatin (1 μ g/ml), and aprotinin (1 μ g/ml). The protein concentrations were determined using Bio-Rad Protein Assay reagent, and equal amounts of protein were subjected to Western blotting. Samples were separated by SDS/PAGE and transferred to polyvinylidene difluoride membranes. After being blocked in Protein-free T20 (TBS) blocking buffer (Fisher Scientific) at room temperature for 1 h, the membranes were rinsed with TBST (0.1%) and incubated at 4°C overnight with primary Abs (1:1000 dilution). The membranes were washed and incubated with secondary Ab (1:5000 dilution) at room temperature for 1 h. Signals were developed using Luminata Classico or Forte HRP substrates (Millipore).

Antibodies

The following Abs acquired from Cell Signaling Technologies were used: TBK1 (3504), p-TBK1 (5483), IKK ϵ (3416), p-IKK ϵ (8766), IRF3 (4302), and p-IRF3 (4947), RIPK1 (3493), α -Tubulin (3873), anti-mouse IgG HRP-linked Ab, and anti-rabbit IgG HRP-linked Ab. RIPK3 Ab was purchased from Prosci (2283). Primary Abs were used at a dilution of 1:1000, and secondary Abs were used at a dilution of 1:5000.

Cell viability assays

The cells were seeded as described earlier in 100 μ l of media. Typically, cells were treated with 10 ng/ml LPS and 50 μ M zVAD for 24 h. Cell viability was determined using CellTiter-Glo viability assay kit (Promega). Each independent experiment was performed in duplicate and repeated three times. Viability of the cells, relative to an untreated control, was determined and plotted.

Measurement of IFN- β by ELISA

Mouse IFN- β was measured using colorimetric ELISA. Ninety-six-well plate was incubated at 4°C overnight after coating wells with 50 μ l of monoclonal rat anti-mouse IFN- β (SC-57201; Santa Cruz) diluted 1:500 (final concentration: 0.2 μ g/ml) in 0.1 M carbonate buffer. The following day, wells were incubated with 200 μ l of blocking buffer (10% FBS in PBS) for 2 h at 37°C. After blocking, wells were incubated with 50 μ l of standard or sample (diluted 1:1, 1:5, and/or 1:25 in blocking buffer) and incubated overnight at 4°C. IFN- β protein purchased from PBL (product number: 12400), and standard curves were generated using concentrations 0–1000 IU/ml (0–2 ng/ml). The following day, wells were incubated with

50 μ l of polyclonal rabbit anti-mouse IFN- β detecting Ab from PBL (product number: 32400) diluted 1:2000 (final concentration: 0.5 μ g/ml) in blocking buffer and incubated overnight at 4°C. The following day, 50 μ l of goat anti-rabbit HRP secondary Ab (Cell Signaling) diluted 1:2000 was added to the wells. Plates were incubated for 2–3 h at room temperature. Subsequently, 50 μ l of the TMB substrate was added. The reaction was stopped by the addition of 50 μ l of 2N H₂SO₄. Absorbance was measured at 450 nm. A washing step with buffer (PBS 0.05% Tween) was performed after each step and before the addition of the substrate.

RNA extraction, cDNA synthesis, and qRT-PCR

For RNA extraction, cells were seeded as described earlier. Cells were stimulated with 10 ng/ml LPS, 50 μ M zVAD, and 30 μ M Nec-1s or other inhibitors as indicated. Total RNA was isolated using RNA MiniPrep kit (ZYMO Research) according to the manufacturer's protocol. Fifty nanograms to 1 μ g of RNA was converted to cDNA using iScript cDNA Synthesis kit (Bio-Rad). One microliter of cDNA was used with 500 pM primers in 20 μ l of quantitative PCRs using VeriQuest SYBR Green master mix (Affymetrix). qRT-PCRs were performed in LightCycler 480 II using the following program: 50°C for 2 min, 95°C for 10 min, followed by 40 cycles of amplification (95°C for 15 s, 60°C for 1 min). GAPDH was analyzed as a housekeeping gene.

The primer sequences used to amplify murine genes are as follows: mouse GAPDH: forward 5'-TGTGTCCGTCGTGGATCTGA-3', reverse 5'-GGTCCTCAGTGTAGCCCAAG-3'; mouse IFN β 1: forward 5'-CAGCTCCAAGA-AAGGACGAAC-3', reverse 5'-GGCAGTGTAACCTCTTCTGCAT-3'; mouse MX2: forward 5'-GAGGCTCTTCAAGATGAGCAA-3', reverse 5'-CTC TGCGGTACGTCTCTCT-3'; and mouse IFIT1: forward 5'-CTGAGATGT-CACTTCACATGGAA-3', reverse 5'-GTGCATCCCCAATGGGTTCT-3'.

Necrosome formation assay

Isolation of Nonidet P-40 (NP-40)-soluble and -insoluble fractions was performed as previously described (15, 43). Cells were seeded into six-well plates at 2×10^6 cells per well and stimulated with 10 ng/ml LPS, 50 μ M zVAD, and 30 μ M Nec-1s for up to 3 h. Cells were lysed in 1% Triton X-100 or 1% NP-40 lysis buffer (150 mM NaCl, 20 mM Tris-Cl [pH 7.5], 1% Triton X-100 or 1% NP-40, 1 mM EDTA, 3 mM Na-fluoride, 1 mM β -glycerophosphate, 1 mM sodium orthovanadate, 5 μ M iodoacetamide, 2 μ M *N*-ethylmaleimide and phosphatase, and 5 μ g/ml PMSF and 1 μ g/ml leupeptin, 1 μ g/ml pepstatin, and 1 μ g/ml aprotinin). Lysates were flash frozen on dry ice, thawed on ice, and vortexed for 10 s followed by centrifugation at $1000 \times g$ for 15 min in the refrigerated tabletop centrifuge to remove nuclei. Supernatants were collected and centrifuged at $34,400 \times g$ for 15 min to precipitate detergent-insoluble cellular fraction (NP-40 or Triton insoluble). Supernatants were collected (NP-40 or Triton-soluble fractions), and pellets were boiled in 1 $\times$ SDS-PAGE buffer.

Sucrose-gradient velocity sedimentation

RAW264.7 macrophages were treated for 3–4 h, and lysates were collected as in the necrosome formation assay. Samples were flash frozen, thawed, and nuclei were precipitated and discarded as in necrosome formation assay. Protein concentration normalized and was placed upon a 10–50% linear sucrose gradient. Samples were subject to velocity sedimentation at $\sim 250,000 \times g$ for 2.5–3 h, and samples were collected in 13–14 $\times$ 1 ml aliquots using a Gilson FC 203B Fraction Collector. Samples were subjected to chloroform/methanol protein precipitation to eliminate sucrose and/or concentrate protein samples before resuspension in 4 $\times$ Laemmli buffer for Western analysis.

Coimmunoprecipitation studies

RAW264.7 macrophages were treated for 3–4 h, and lysates were collected as in the necrosome formation assay. Samples were vortexed and incubated on ice before precipitating nuclei. Lysate protein concentration was normalized to 2 mg/ml across samples, and lysates were incubated overnight with rabbit anti-RIPK1 or rabbit anti-IKK ϵ Ab. Ab-bound protein was captured using Protein A-conjugated magnetic beads per sample (Pierce Protein A Magnetic Beads 88845; ThermoFisher Scientific) and eluted by boiling in 1 $\times$ Laemmli buffer for 5 min.

RNA sequencing

For RNA-sequencing analysis, mice were divided into three groups: control ($n = 2$), LPS ($n = 2$), and LPS/Nec-1s ($n = 2$). Bone marrow cells were isolated by FACS as described for quantitative PCR analysis. Total RNAs were isolated using Qiagen RNeasy kit according to the manufacturer's protocol. Input RNA samples were first analyzed by Agilent Bioanalyzer 2100 to assess the integrity and quantity. The RNA samples were then

amplified using NuGen Ovation RNA System V2. The resulting cDNA samples were fragmented on Covaris M220 Focused Sonicator, followed by purification and concentration with a Qiagen MiniElute Spin Column. After this step, S1 Nuclease (Promega) was used according to the manufacturer's protocol. Then, the amplified and fragmented cDNA samples from RNA amplification were used as input for library preparation, using Illumina TruSeq DNA Sample Preparation Kit per the manufacturer's instruction. The resulting libraries were quantified and pooled at equal molar concentration for sequencing. The sequencing was done on a lane of high-output single-read 100 bases on an Illumina HiSeq2500 using SBS V3 chemistry. The base calling and demultiplexing were performed with CASAVA v1.8. The resulting data were then aligned to mouse mm10 reference genome with Tophat 2 and different gene expression analysis with Cuffdiff. Gene expression profiles were analyzed using Ingenuity Pathway Analysis software to identify pathways regulated by LPS and Nec-1s. The sequences used are found in Gene Expression Omnibus (<http://www.ncbi.nlm.nih.gov/geo/>) under accession number GSE73836.

Statistics

For in vivo studies, statistics were analyzed by two-tailed Student *t* test, and significance was determined using an α value of 0.05. Error bars reflect SEM. For in vitro studies, experiments were repeated with at least two, but in most cases three, biological replicates. In each figure, one representative dataset is shown due to the variability in the general responsiveness to LPS in independent cell preparations. Error bars reflect SD from the mean with measurements made in duplicate.

Results

Kinase activity of RIPK1 is required for IFN- β synthesis induced by LPS in mice

In a previous study, we reported that kinase activity of RIPK1 was required for acute inflammatory cytokine production induced by LPS in mice, including TNF- α , IL-6, CCL3/4, and CXCL1/2, suggesting that the catalytic activity of RIPK1 might serve an important role in the innate immune response induced by TLR4 (14). Synthesis of IFN-I, including IFN- β (44, 45), is elicited by TLR4 activation and plays an important role in innate and adaptive immunity against infections (46). To inquire whether the catalytic activity of RIPK1 may also be required for LPS-induced IFN- β synthesis, we first tested the effect of the selective RIPK1 inhibitor, Nec-1s, on this process. Nec-1s injection before a challenge with LPS abolished IFN- β mRNA synthesis in CD11b⁺ monocytic cells in vivo (Fig. 1A). In addition, we observed that LPS-driven induction of a panel of IFN-stimulated genes was similarly abolished by Nec-1s in CD11b⁺ cells (Fig. 1B). Similarly, LPS-induced IFN- β synthesis was also attenuated in monocytic CD11b⁺ cells from D138N RIPK1 kinase-inactive mice (D138N RIPK1 or *Ripk1*^{D138N/D138N}) (Fig. 1C). Together, these pharmacologic and genetic approaches suggest that RIPK1 kinase is indeed required for LPS-induced upregulation of IFN- β expression in vivo.

LPS with zVAD.fmk induces RIPK1 kinase-dependent IFN- β synthesis in BMDMs

Deletion of caspase-8 or use of the pan-caspase inhibitor, zVAD, is sufficient to facilitate RIPK1 kinase activation by LPS in vitro (9, 12, 14, 47). Within 24 h, LPS and zVAD treatment, but not LPS alone, induced RIPK1 kinase-dependent cell death in BMDMs (Supplemental Fig. 1A, 1B). Requirement for caspase inhibition for RIPK1 activation in vitro is also supported by our previously published phospho-RIPK1 ELISA data (14). To address the possible role of RIPK1 in IFN-I production, we first examined whether zVAD promoted LPS-induced IFN- β synthesis in BMDMs. Indeed, LPS paired with zVAD greatly increased synthesis of IFN- β compared with LPS alone over an 8-h time course (Fig. 2A). Similarly, mRNA expression of ISGs, MX2 and IFIT1, was augmented by LPS with zVAD compared with LPS alone (Fig. 2B, 2C).

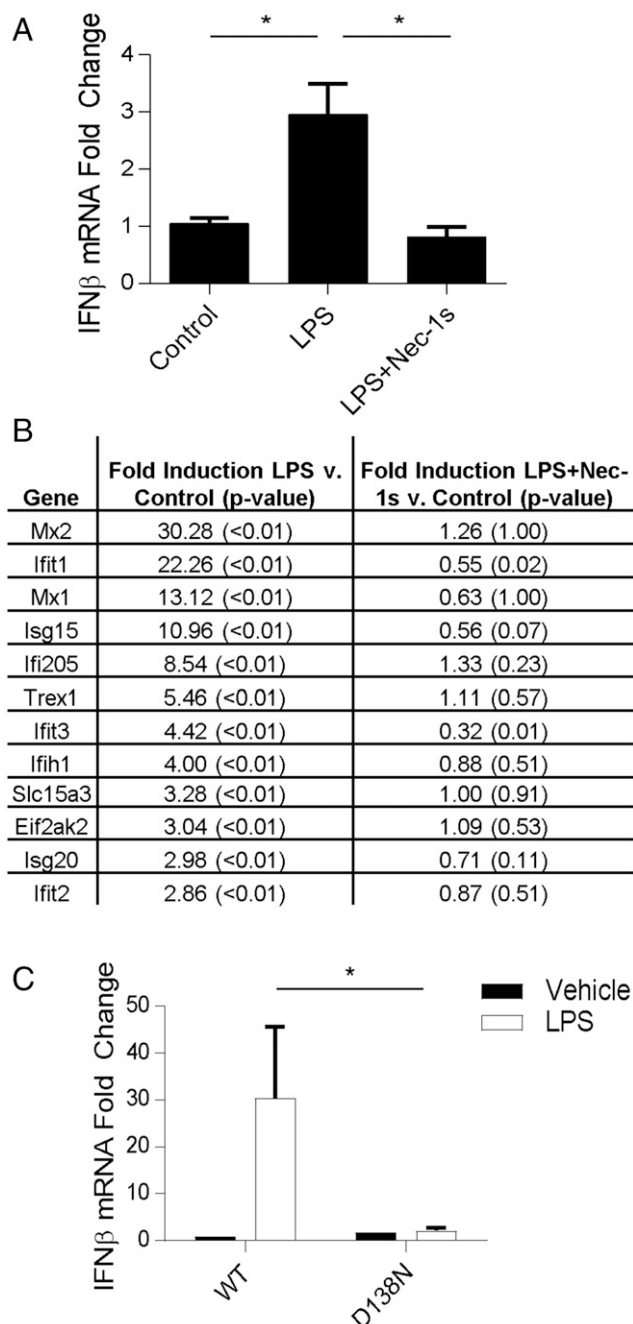


FIGURE 1. LPS-induced IFN-I production is dependent on kinase activity of RIPK1. **(A)** qRT-PCR analysis of *Ifnb* mRNA expression in WT mice injected with Nec-1s (i.v.) 15 min before LPS (i.p.). $n = 6$ animals per group. $*p < 0.05$. Values reflect mean $\pm$ SE. **(B)** RNA sequencing analysis of a panel of IFN-I response genes in CD11b $^{+}$ cells isolated from mice injected as in (A). Values reflect mean. The p values are marked in parentheses. RNA sequencing data, accession number GSE73836. **(C)** qRT-PCR analysis of *Ifnb* mRNA expression on WT and D138N mice injected with LPS (i.p.). $n = 3$ –7 animals per group. $*p < 0.05$. Values reflect mean across biological variants, and error bars reflect SE.

Next, to evaluate whether this regulation was dependent on the catalytic activity of RIPK1 kinase, we examined IFN- β synthesis in BMDMs generated from two different mouse strains expressing kinase-dead mutants of RIPK1 (D138N RIPK1 or K45A RIPK1 [*Ripk1*^{K45A/K45A}]) mice (33, 34). Consistent with our hypothesis, we observed that in the absence of the kinase activity of RIPK1, increases in IFN- β mRNA synthesis and protein release in response to LPS with zVAD was abolished or attenuated, respectively

(Fig. 2D–G). Nec-1s also blocked IFN- β production by LPS with zVAD (Fig. 2D–G). Conversely, catalytic activity of RIPK1 was not essential for the responses to LPS alone. Together, these data suggest that activation of RIPK1 in the presence of caspase inhibition greatly promotes IFN- β synthesis in BMDMs.

IFN- β synthesis induced by LPS with zVAD requires kinase activity of RIPK3, but not MLKL

RIPK3 is a central component of the necroptosis signaling cascade and is required for necroptosis induced by LPS with zVAD (48). Catalytic activity of RIPK3 is required for the formation of detergent-insoluble RIPK1-RIPK3 necrosome complexes and consequent phosphorylation of MLKL, which ultimately leads to the execution of necroptosis (4, 14, 17, 19, 20, 23, 49). Consistently, we observed that K51A RIPK3 BMDMs (*Ripk3*^{K51/K51A}), expressing low levels of a kinase-dead mutant of RIPK3 (39), and RIPK3 knockout (*Ripk3*^{-/-}) BMDMs were protected from cell death induced by LPS with zVAD (Supplemental Fig. 2A, 2B).

We next sought to determine whether RIPK1 kinase-dependent IFN- β production required these additional necroptotic factors. Stimulation of RIPK3 K51A BMDMs by LPS with zVAD failed to induce IFN- β production above the response by LPS alone, indicating that RIPK3 was required for RIPK1 kinase-dependent IFN- β production (Fig. 3A). To further evaluate the role of catalytic activity of RIPK3, we used the specific RIPK3 kinase inhibitor GSK'872 (9, 39). We observed that GSK'872 blocked IFN- β synthesis induced by LPS with zVAD, confirming that the catalytic activity of RIPK3 is also required for RIPK1 kinase-dependent IFN- β production in BMDMs (Fig. 3B). In contrast, deletion of *Mkl1* only minimally impaired IFN- β mRNA synthesis induced by LPS with zVAD, even though these BMDMs were, expectedly, completely resistant to necroptosis (Fig. 3C, 3D) (14). These observations demonstrated that RIPK1 and RIPK3 kinase-dependent IFN- β synthesis is independent of the necroptosis pathway, and that bifurcation of the respective signaling cascades occurs upstream of MLKL. Furthermore, given the paucity of known targets of RIPK1 and RIPK3, these findings highlight the possibility of new cell death-independent enzymatic targets for these kinases that remain to be defined.

IFN- β production by LPS with zVAD requires TRIF, STING, TBK1/IKK ϵ , and IRF3/7

IFN- β production, induced by LPS alone, is dependent on a number of key intermediaries starting with the adapter protein TRIF, which mediates signaling through canonical downstream IFN-I pathway intermediaries, including homologous kinases TBK1 and IKK ϵ , and the transcription factors, IRFs (50–53). Because TRIF (*Ticam1*) is required for RIPK1- and RIPK3-dependent necroptosis (13), we evaluated IFN- β mRNA synthesis in *Ticam1*^{-/-} BMDMs to examine whether TRIF is similarly required for the RIPK1 and RIPK3 kinase-dependent IFN- β synthesis. Not surprisingly, in the absence of TRIF, neither LPS nor LPS with zVAD induced IFN- β mRNA synthesis (Fig. 4A). Therefore, next we sought to determine whether increased IFN- β synthesis upon RIPK1 and RIPK3 activation reflects induction of the canonical IFN-I axis (TBK1, IKK ϵ , and IRF3) or whether a new TRIF-dependent signaling mechanism is engaged by RIPK1 and RIPK3. Over an extended time course, we found that LPS with zVAD enhanced phosphorylation of TBK1, IKK ϵ , and IRF3 compared with LPS alone, suggesting IFN-I pathway hyperactivation (Fig. 4B, Supplemental Fig. 3A–C). In addition, TBK1, IKK ϵ , and IRF3 phosphorylation in the cells stimulated with either LPS alone or LPS with zVAD were reduced in *Ticam1*^{-/-} BMDMs, confirming that TRIF was required for IFN-I pathway

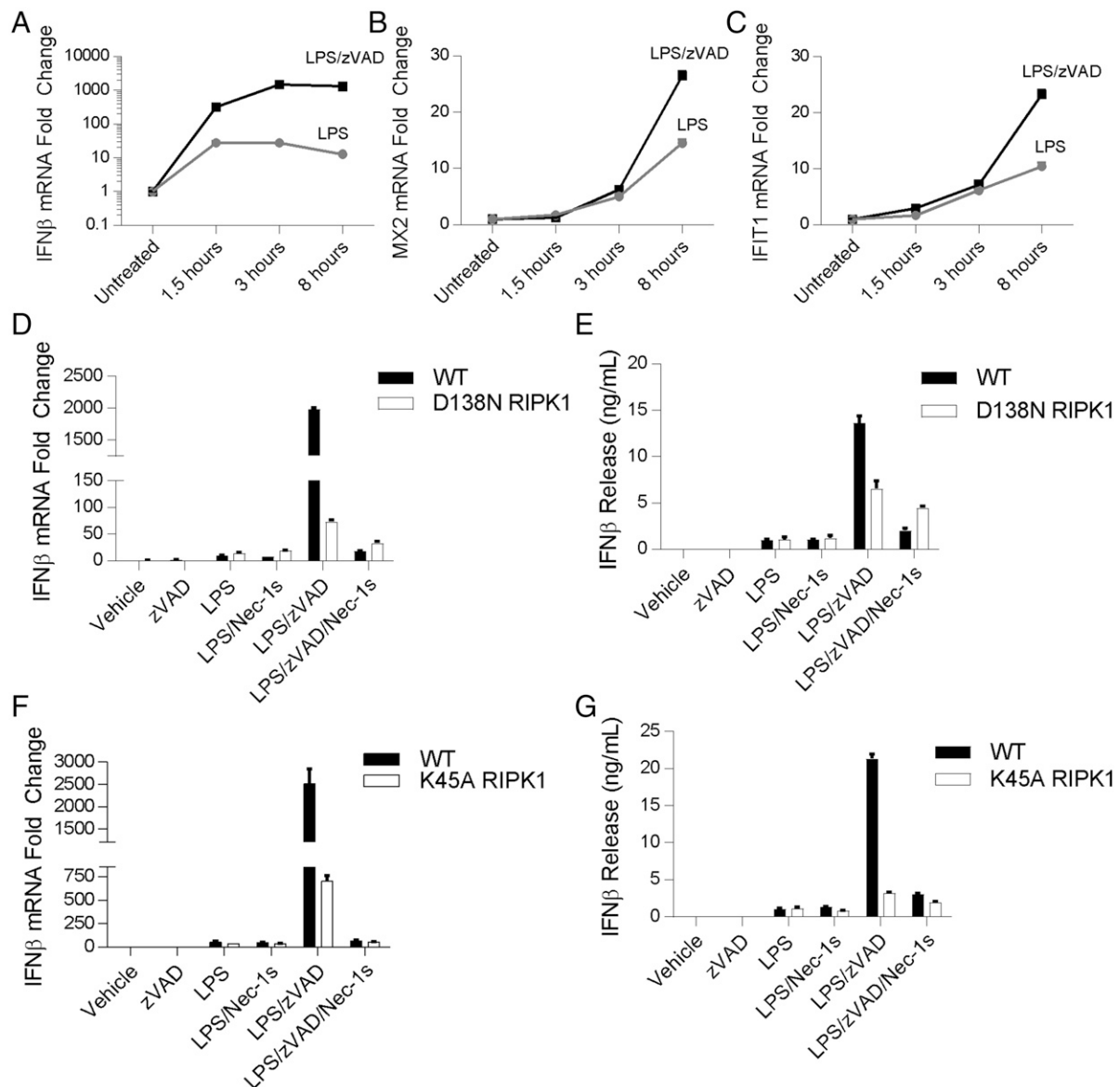


FIGURE 2. LPS with zVAD induces IFN- β synthesis in a RIPK1 kinase-dependent manner. (**A–C**) Time course of mRNA expression changes in *Ifnb* (**A**), *Mx2* (**B**), and *Ifit1* (**C**) evaluated by qRT-PCR in WT BMDMs. Black squares represent LPS/zVAD; gray circles represent LPS. (**D** and **E**) qRT-PCR and ELISA analysis of *Ifnb* mRNA expression and IFN- β protein release in WT and D138N BMDMs treated for 5–7 h. (**F** and **G**) qRT-PCR and ELISA analysis of *Ifnb* mRNA expression and IFN- β protein release in WT and K45A BMDMs treated for 5–7 h. Representative data are shown. Error bars reflect SD from the mean. BMDMs were treated with LPS = 10 ng/ml, zVAD = 50 μ M, and/or Nec-1s = 30 μ M where indicated.

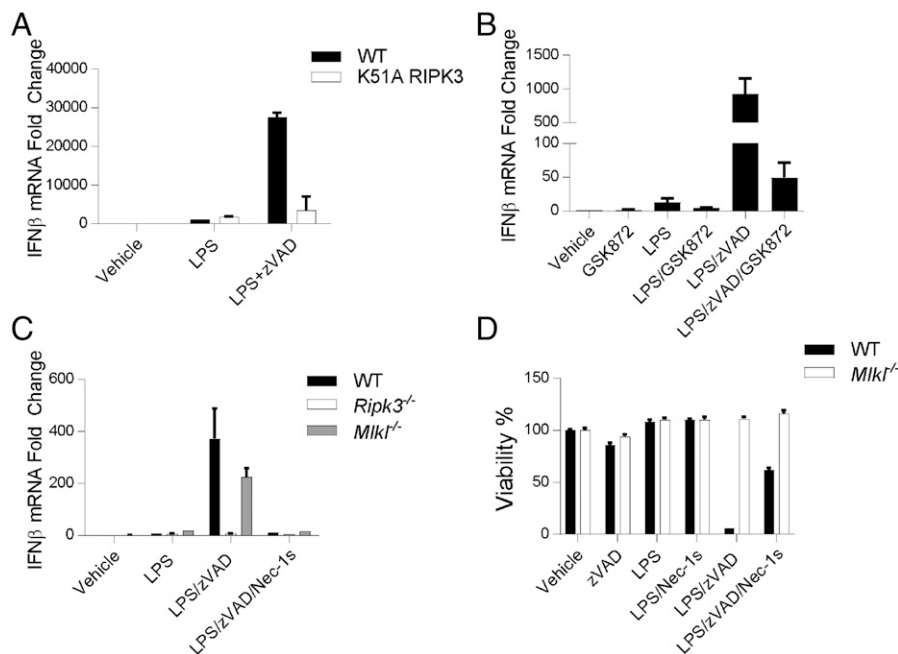
activation by LPS and zVAD (Fig. 4C). By comparison, early TRIF-dependent activation of the IFN-I pathway (1 h posttreatment) was not affected by zVAD or blocked by Nec-1s, suggesting that the catalytic function of RIPK1 is not required for early signaling events in this pathway (Fig. 4D). A recent study demonstrated that TRIF directly associated with STING, an intracellular nucleotide sensor, and was also required for STING-dependent IFN-I synthesis (54). Consistent with a role for STING as a codriver for TRIF-dependent IFN- β synthesis, we found that STING was also required for TRIF/RIPK1-dependent IFN- β synthesis by LPS with zVAD (Fig. 4E).

To test the functional roles of TBK1 and IKK ϵ in RIPK1 and RIPK3 kinase-dependent IFN- β synthesis, two dual TBK1/IKK ϵ inhibitors (MRT67307 and BX795) were used (40, 41). These inhibitors blocked both LPS and LPS with zVAD-induced IFN- β mRNA synthesis (Fig. 4F, Supplemental Fig. 3D), and we confirmed that MRT67307 also blocked phosphorylation of IRF3 induced by LPS and LPS with zVAD (Fig. 4G).

To examine the contribution of IRFs to RIPK1 and RIPK3 kinase-dependent IFN- β production, we evaluated responses in *Irf3*^{−/−}/*Irf7*^{−/−} BMDMs. Again, IFN- β mRNA synthesis induced by LPS or LPS with zVAD were completely abolished in the absence of IRF3 and IRF7, demonstrating that IRFs are also required for RIPK1 and RIPK3 kinase-dependent IFN- β synthesis (Fig. 4H). Altogether, these data suggest that TRIF plays a dual role in promoting signaling by both RIPK1 and RIPK3 kinases and TBK1/IKK ϵ , as well as allowing cross-talk between the two pathways to enhance activation of the TBK1, IKK ϵ , and IRF3 signaling cascade.

Lastly, we examined whether activation of IFN-I pathway intermediaries was also regulated by catalytic activities of RIPK1 and RIPK3. Importantly, phosphorylation of TBK1, IKK ϵ , and IRF3 was attenuated in K45A RIPK1 BMDMs and D138N RIPK1 BMDMs and inhibited by Nec-1s in wild-type (WT) BMDMs, confirming the requirement for RIPK1 kinase in this regulation

FIGURE 3. Kinase activity of RIPK3, but not MLKL, is required for RIPK1 kinase-dependent IFN- β synthesis. **(A)** qRT-PCR analysis of *Ifnb* mRNA expression in WT and K51A BMDMs treated for 5–7 h. **(B)** qRT-PCR of *Ifnb* mRNA expression in WT BMDMs treated with RIPK3 kinase inhibitor (GSK872 = 5 μ M) for 5–7 h. **(C)** qRT-PCR of *Ifnb* mRNA expression in WT, RIPK3 knockout (*Ripk3*^{-/-}), and MLKL knockout (*Mkl*^{-/-}) BMDMs treated for 7 h. **(D)** Cell viability of WT and *Mkl*^{-/-} BMDMs treated for 24 h and evaluated by CellTiter-Glo ATP assay. Representative data are shown. Error bars reflect SD from the mean. BMDMs were treated with LPS = 10 ng/ml, zVAD = 50 μ M, GSK'872 = 5 μ M, and/or Nec-1s = 30 μ M where indicated.



(Fig. 5A, 5B). Similarly, these events were abolished in K51A RIPK3 BMDMs, in *Ripk3*^{-/-} BMDMs, and by GSK'872 (Fig. 5C, 5D, Supplemental Fig. 4). Notably, RIPK1 kinase was specifically required for TRIF-dependent signaling events that were augmented by LPS with zVAD because early activation of TBK1, IKK ϵ , and IRF3 1 h posttreatment did not require RIPK1 kinase (Fig. 4D). Together, these observations suggest that RIPK1 and RIPK3 kinase-dependent IFN- β synthesis is mediated by the canonical TBK1/IKK ϵ /IRF signaling cascade and that this regulation is also dependent on the adapter TRIF.

LPS with zVAD promotes localization and activation of TBK1, IKK ϵ , and IRF3 in RIPK1- and RIPK3-containing detergent-insoluble cellular fractions

Observing that IFN-I pathway intermediaries were required for RIPK1 and RIPK3 kinase-dependent IFN- β synthesis, we considered the possibility that these intermediaries associate with RIPK1 and RIPK3 necrosomes. Activation of RIPK1 and RIPK3 kinase-dependent cell death is associated with posttranslational modification and enrichment of the kinases in detergent-insoluble cellular fractions, representing necrosome complexes (14, 15, 43, 55). LPS and zVAD treatment of BMDMs resulted in the coenrichment of modified forms of RIPK1 and RIPK3 in detergent-insoluble cellular fractions that were blocked by Nec-1s (Fig. 6A), consistent with our previous report of the formation of detergent-insoluble necrosome-like aggregates in LPS- and zVAD-treated BMDMs (14). We next analyzed coenrichment of IFN-I pathway intermediaries with RIPK1 and RIPK3 under these conditions. Notably, we observed that TBK1, IKK ϵ , and IRF3 were enriched in detergent-insoluble fractions (Fig. 6B). Moreover, we found that phosphorylated or activated forms of these intermediaries were enriched in these cellular compartments (Fig. 6B). This regulation was dependent on the catalytic function of RIPK1 and RIPK3 because enrichment of these factors was abolished in the presence of Nec-1s or in K51A RIPK3 BMDMs (Fig. 6B, 6C). We further examined coenrichment with RIPK1 and RIPK3 using additional high m.w. protein complex fractionation by sucrose gradient-based velocity sedimentation. These experiments were performed in an immortalized macrophage cell line (RAW264.7

cells), providing a more abundant source of material than BMDMs, using an optimized pan-caspase inhibitor, IDN6556 (42). RIPK1 and RIPK3 were enriched in fractions 12 and 13, indicating the formation of a high m.w. complex, when cells were treated by LPS and a pan-caspase inhibitor that was abolished in the presence of Nec-1s (Fig. 6D). Consistent with the previous data, TBK1, IKK ϵ , and IRF3 were again coenriched in these same fractions (Fig. 6D). Intriguingly, additional coimmunoprecipitation studies revealed that TBK1 and IKK ϵ directly complexed with RIPK1, as did RIPK3, when cells were treated with LPS with IDN, and this was abolished in the presence of Nec-1s (Fig. 6E). We also found that TRIF association with IKK ϵ was not modified in a RIPK1 kinase-dependent manner, suggesting that RIPK1 kinase-dependent signaling may be mediated by posttranslational modification as opposed changes in molecular association (Fig. 6F). Lastly, to confirm that RIPK1 and RIPK3 kinase-dependent IFN-I pathway signaling did not also require MLKL, we examined enrichment of IFN-I pathway intermediaries in detergent-insoluble fractions in *Mkl*^{-/-} BMDMs. Consistent with our previous observation that MLKL was not required for RIPK1 and RIPK3 kinase-dependent IFN- β synthesis, we observed that MLKL was also dispensable for coenrichment and activation of TBK1, IKK ϵ , and IRF3 in necrosome-containing cellular fractions (Fig. 6G). Together, these data support the model that IFN-I pathway intermediaries may be recruited and directly activated by RIPK1 and RIPK3 necrosome complexes.

RIPK1 and RIPK3 kinase-dependent IFN- β synthesis in macrophages is induced by avirulent strains of Gram-negative bacteria

Several studies have demonstrated that TRIF-dependent IFN- β production is an important feature of the host response that aids in the resolution of Gram-negative bacterial infections (56–58). However, Gram-negative bacteria pathogens such as *Yersinia* and *Klebsiella* have evolved an armament of virulence factors to manipulate host responses and facilitate their pathogenesis. In the case of *Yersinia*, type III secretion system Yop proteins are injected directly into host cells (59–61). Several of these Yops impede host-cell defenses by blocking host proinflammatory signaling. Similarly, *Klebsiella* is equipped with an outer polysaccharide

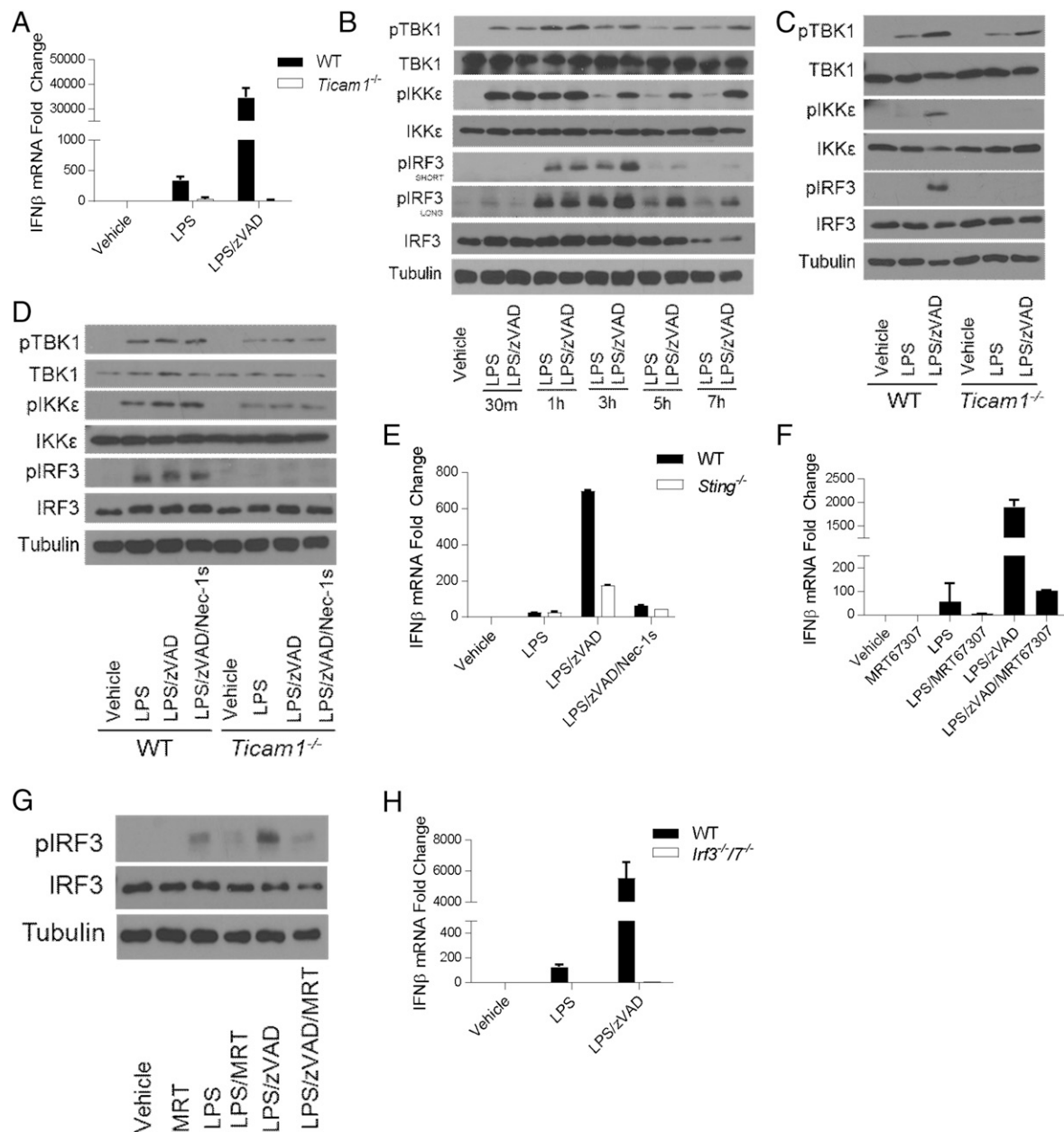


FIGURE 4. RIPK1 kinase-dependent IFN- β synthesis requires TRIF, STING, TBK1/IKK ϵ , and IRF3/7. (A) qRT-PCR of *Ifnb* mRNA expression in WT and *Ticam1*^{-/-} BMDMs treated for 5–7 h. (B) Time course of TBK1, IKK ϵ , and IRF3 phosphorylation evaluated by Western analysis in WT BMDMs. (C) Western analysis of TBK1, IKK ϵ , and IRF3 phosphorylation in WT and *Ticam1*^{-/-} BMDMs treated for 3–4 h. (D) Western analysis of TBK1, IKK ϵ , and IRF3 phosphorylation in WT and *Ticam1*^{-/-} BMDMs treated for 1 h. (E) qRT-PCR analysis of *Ifnb* mRNA expression in WT and *Sting*^{-/-} BMDMs treated for 5–7 h. (F) qRT-PCR of *Ifnb* mRNA expression in WT BMDMs treated with TBK1/IKK ϵ inhibitor (MRT67307, 2 μ M) for 5–7 h. (G) Western analysis of IRF3 phosphorylation in WT BMDMs treated with MRT67307 for 3–4 h. (H) qRT-PCR of *Ifnb* mRNA expression in WT and *Irf3*^{-/-} BMDMs treated for 5–7 h. Representative data are shown. Error bars reflect SD from the mean. BMDMs were treated with LPS = 10 ng/ml, zVAD = 50 μ M, Nec-1 = 30 μ M, and/or MRT67307 = 2 μ M where indicated.

capsule that facilitates immune evasion and enhances bacterial virulence (62, 63). We evaluated whether these pathogens use their virulence factors to dampen sensing of their LPS and subsequent IFN- β induction by RIPK1 and RIPK3.

We examined IFN- β mRNA synthesis in the presence of zVAD postinfection with WT or avirulent forms of *Y. pseudotuberculosis* and *K. pneumoniae*. Notably, the avirulent strain of *Y. pseudotuberculosis*, lacking the ability to inject Yops into host cells (Δ yscF), and the avirulent strain of *K. pneumoniae* (Δ cpsB), unable to produce an outer polysaccharide capsule, induced a robust increase in IFN- β mRNA synthesis compared with WT or path-

ogenic counterparts in the presence of zVAD (Fig. 7A, 7B). To verify that IFN- β induction was dependent on RIPK1 and RIPK3 kinases, we also determined that the response was absent in K45A RIPK1 and K51A RIPK3 BMDMs (Fig. 7A, 7B). Furthermore, consistent with the previous data, we again observed robust RIPK1 kinase activity-dependent coaccumulation of RIPK1, RIPK3, and IFN-I pathway components in the insoluble fractions of cells infected with Δ yscF or Δ cpsB in the presence of zVAD (Fig. 7C, 7D). In sum, these data provide direct demonstration that Gram-negative bacteria are capable of inducing RIPK1 and RIPK3 kinase-dependent IFN- β synthesis in vitro in the presence of zVAD, and that this induction may be

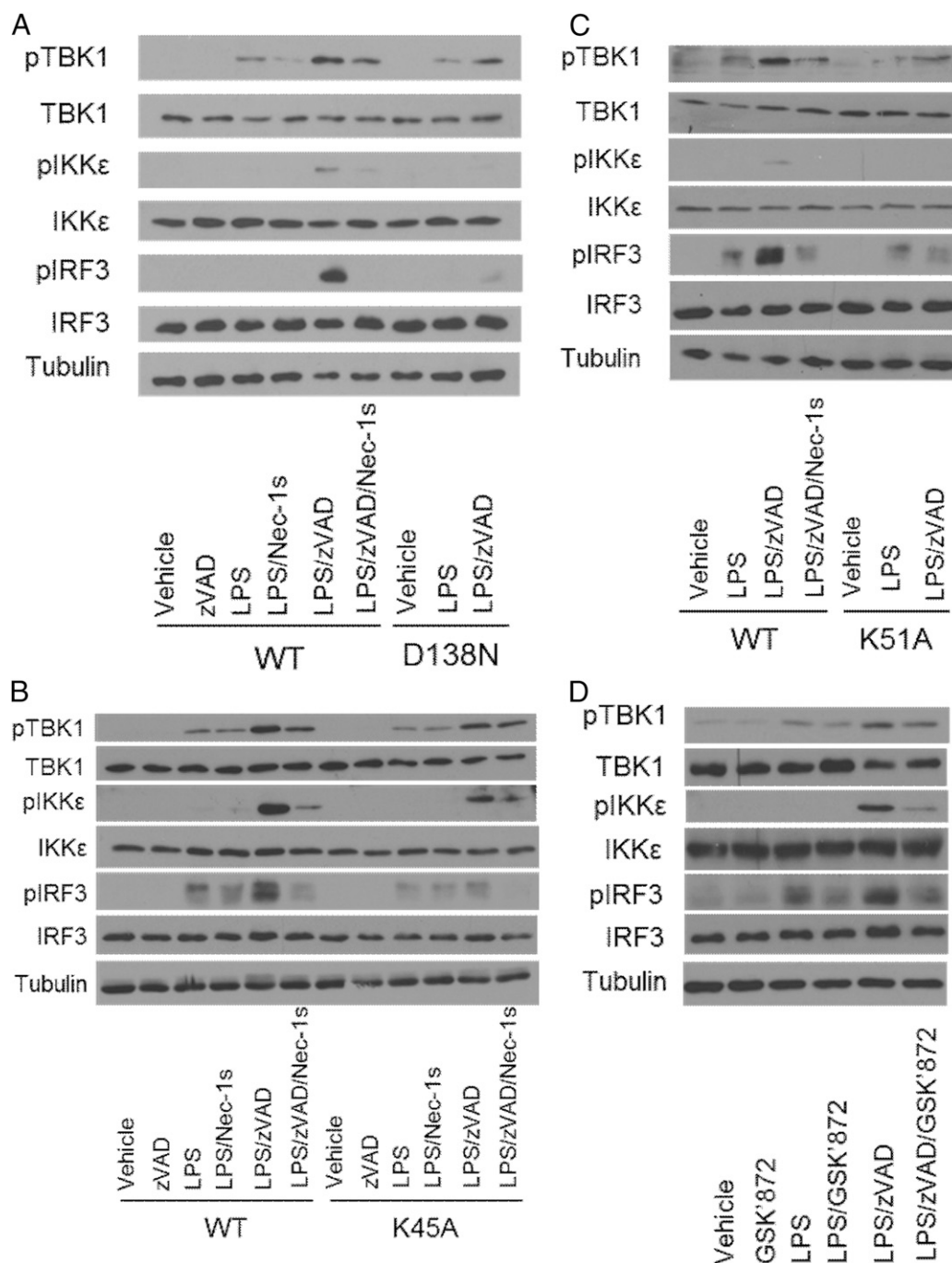


FIGURE 5. IFN-I pathway activation by LPS with zVAD is RIPK1 and RIPK3 kinase dependent. **(A)** Western analysis of TBK1, IKK ϵ , and IRF3 phosphorylation in WT and D138N RIPK1 (D138N) BMDMs treated for 3–4 h. **(B)** Western analysis of TBK1, IKK ϵ , and IRF3 phosphorylation in WT and K45A RIPK1 (K45A) BMDMs treated for 3–4 h. **(C)** Western analysis of TBK1, IKK ϵ , and IRF3 phosphorylation in WT and K51A RIPK3 (K51A) BMDMs treated for 3–4 h. **(D)** Western analysis of TBK1, IKK ϵ , and IRF3 phosphorylation in WT BMDMs treated with RIPK3 kinase inhibitor (GSK'872) for 3–4 h. BMDMs were treated with LPS = 10 ng/ml, zVAD = 50 μ M, Nec-1 = 30 μ M, and/or GSK'872 = 5 μ M where indicated.

under negative regulation by bacterial virulence factors. Accordingly, these data, paired with our *in vivo* observations of LPS-induced RIPK1 kinase-dependent IFN- β synthesis, suggest a potential role for RIPK1 kinase-dependent IFN- β synthesis as part of a physiologic host response in bacterial infection (Fig. 8).

Discussion

RIPK1 and RIPK3 activation and necrosis are generally believed to be associated with acute and pathologic injury; however, emerging evidence suggests RIPK1 and RIPK3 may play significant roles in

physiologic host-defense responses to viral and bacterial infections (64, 65). To date, studies have focused on the roles of RIPK1 and RIPK3 in the context of pathogen-induced cell death. For example, RIPK1 and caspase-8-dependent apoptosis has been described as a feature of the innate immune response to Gram-negative bacterial pathogen, *Yersinia* (66, 67). Similarly, RIPK1 and RIPK3 kinase-dependent necroptosis have been implicated in macrophage cell death by *Salmonella* and local tissue damage in necrotizing pneumonia caused by methicillin-resistant *Staphylococcus aureus* (68, 69). However, inflammatory signaling

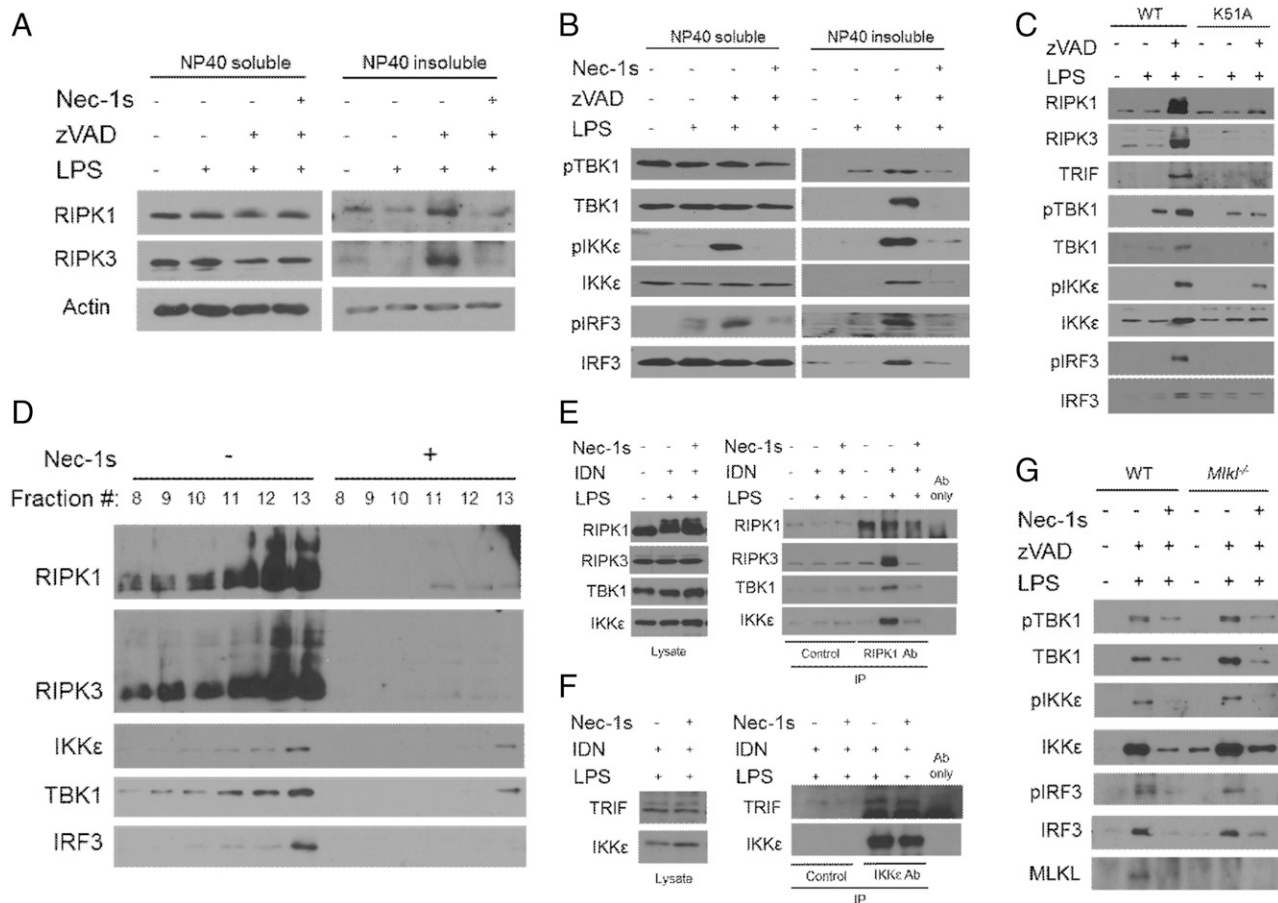


FIGURE 6. Kinase activity of RIPK1 and RIPK3 are required for localization and activation of IFN-I pathway intermediaries in detergent-insoluble cellular fractions. **(A)** Western analysis of localization and modification of RIPK1 and RIPK3 in NP-40-soluble and NP-40-insoluble fractions from WT BMDMs treated for 3–4 h. **(B)** Western analysis of localization and phosphorylation of TRIF, TBK1, IKKε, and IRF3 in NP-40-soluble and NP-40-insoluble fractions from WT BMDMs treated for 3–4 h. **(C)** Western analysis of RIPK1, RIPK3, TRIF, TBK1, IKKε, and IRF3 in NP-40-insoluble fractions from WT and K51A RIPK3 (K51A) BMDMs treated for 3–4 h. **(D)** Western analysis of RIPK1, RIPK3, TBK1, IKKε, and IRF3 in RAW264.7 macrophages treated with LPS and IDN6556 (10 μM) ± Nec-1s for 3–4 h. Lysates were fractionated on a linear sucrose gradient by velocity sedimentation, and protein was collected after chloroform-methanol precipitation. **(E)** Coimmunoprecipitation analysis of RIPK3, TBK1, and IKKε after immunoprecipitation of RIPK1 from RAW264.7 macrophages treated for 3–4 h. **(F)** Coimmunoprecipitation analysis of TRIF after immunoprecipitation of IKKε from RAW264.7 macrophages treated for 3–4 h. Control = beads only. **(G)** Western analysis of RIPK1, RIPK3, TRIF, TBK1, IKKε, and IRF3 in NP-40-insoluble fractions from WT and *Mkl1*^{−/−} BMDMs treated for 3–4 h. BMDMs were treated with LPS = 10 ng/ml, zVAD = 50 μM, IDN6556 = 10 μM, and/or Nec-1s = 30 μM where indicated.

is also a critical component of the initial innate immune response and has not been as closely examined in RIPK1 and RIPK3 biology.

Previous work suggested that kinase activities of RIPK1 and RIPK3 may not contribute to cytokine synthesis downstream of TLR3 and TLR4 (7, 8, 32). However, these studies were performed in the absence of caspase inhibition, which is a requisite to elicit kinase functions of RIPK1 and RIPK3 in vitro (9, 12–14). In contrast, data in this study, as well as other recent reports (14, 24, 27, 70), describe RIPK1 and RIPK3 kinase-dependent cytokine responses that manifest independent of MLKL-dependent cell death signaling. In a prior study, we reported that RIPK1 and RIPK3 kinase-dependent cytokine profile evaluated in vitro, in the presence of zVAD, correlated strongly with the inflammatory profile observed in vivo, in the absence of exogenous caspase manipulation (14). We observed that the dichotomy between RIPK1 kinase regulation in vitro and in vivo was linked to limited caspase-8 activation in bone marrow monocytic cells collected from mice injected with LPS (14). Similarly, in this study we find that exogenous inhibition of caspases is not requisite for RIPK1 and RIPK3 kinase-dependent IFN-β synthesis in vivo after LPS

challenge. These findings suggest that use of zVAD in vitro may aid in uncovering new proinflammatory roles for RIPK1 and RIPK3 that manifest in vivo.

In vitro observation of detergent-insoluble, amyloid-like RIPK1 and RIPK3 complexes, also known as necrosomes, have been linked to RIPK1 and RIPK3 kinase-dependent signaling (14, 15, 43). Indeed, it is well established that caspase inhibition is essential to the induction of necroptosis and that MLKL is recruited to necrosome signaling platforms (9, 12–14). Similarly, in the presence of caspase inhibition, we also observe enrichment and activation of IFN-I pathway intermediaries in necrosome-containing cellular fraction, and remarkably, this did not require MLKL. Although the necrosome was originally defined as a cell death signaling complex, our data in this study and in our previous study suggest that the necrosome may additionally serve as a signaling platform for the induction of IFN-β and other cytokines (14). It is important to note that the precise details of necrosome signaling are poorly understood, and it remains to be determined whether a singular necrosome platform is capable of driving cytokine and prodeath signaling independent of one another. Alternatively, the

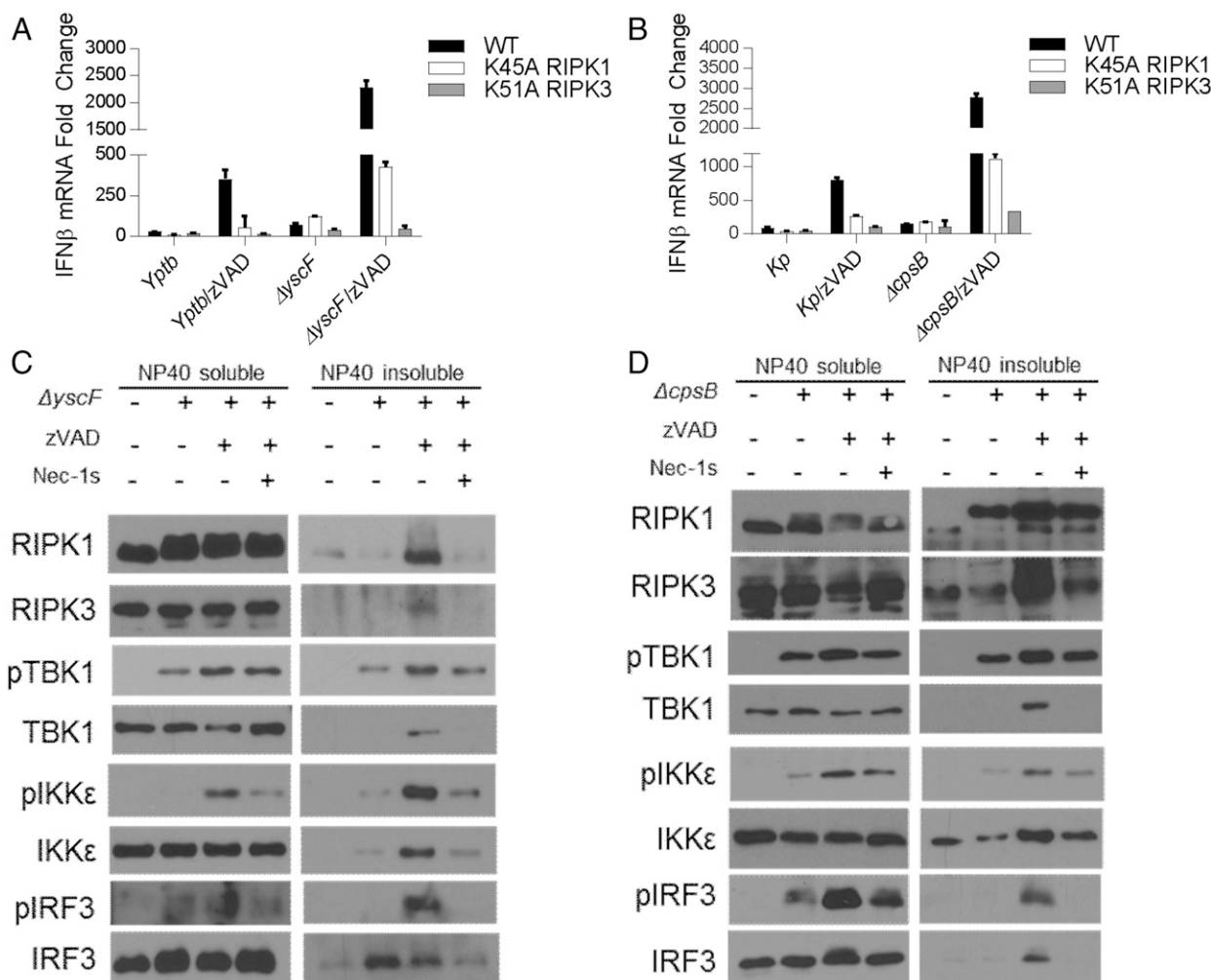


FIGURE 7. RIPK1 and RIPK3 kinase-dependent IFN- β synthesis is augmented by attenuated strains of *Y. pseudotuberculosis* and *K. pneumoniae*. **(A)** qRT-PCR of *Ifnb* mRNA expression in WT, K45A RIPK1 (K45A), and K51A RIPK3 (K51A) BMDMs infected with WT *Y. pseudotuberculosis* or a mutant strain unable to inject pathogenicity factors into macrophages ($\Delta yscF$), at an MOI of 40–60. **(B)** qRT-PCR of *Ifnb* mRNA expression in WT, K45A, and K51A BMDMs infected with WT *K. pneumoniae*, or a mutant strain lacking its out polysaccharide capsule virulence factor ($\Delta cpsB$) at an MOI of 40–60. **(C)** Western analysis of localization and/or phosphorylation of RIPK1, RIPK3, TBK1, IKK ϵ , and IRF3 in NP-40-soluble and NP-40-insoluble fractions from WT BMDMs treated as described in (A). **(D)** Western analysis of localization and/or phosphorylation of RIPK1, RIPK3, TBK1, IKK ϵ , and IRF3 in NP-40-soluble and NP-40-insoluble fractions from WT BMDMs treated as described in (B). Representative data are shown. Error bars reflect SD from the mean. BMDMs were treated with zVAD = 50 μ M and/or Nec-1s = 30 μ M where indicated. Gentamicin (100 μ g/ml) was added 2 h postinfection in each experiment.

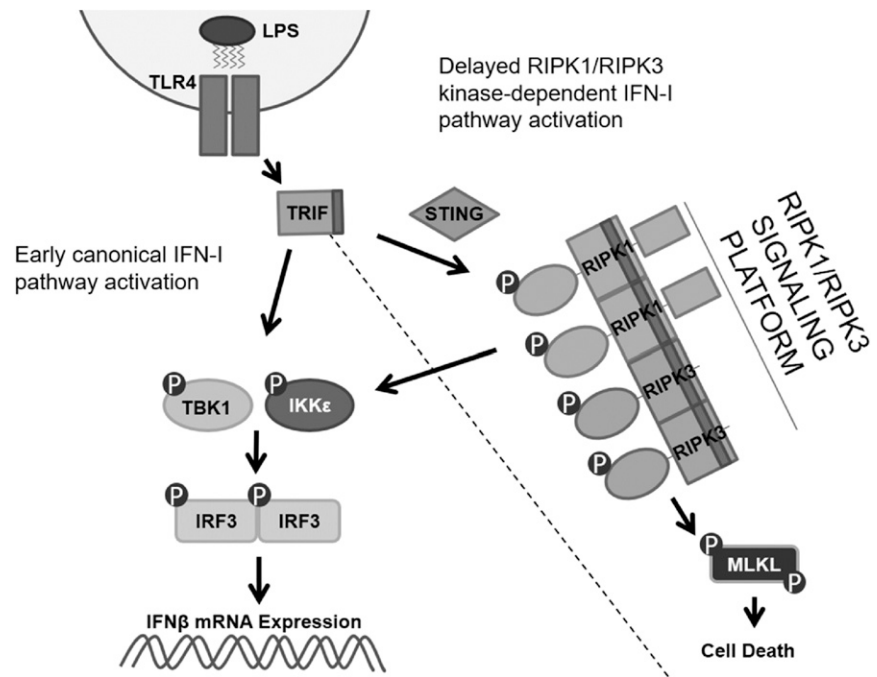
necrosome might reflect physically and/or compositionally heterogeneous RIPK1- and RIPK3-containing platforms that are defined and influenced by their immediate molecular context to promote cytokine synthesis or prodeath signaling.

Our data found that STING contributes to TRIF/RIPK1-dependent IFN- β synthesis (Fig. 4E). Recent work by Wang et al. (54) suggested that TRIF is required for STING to activate IFN- β synthesis. Thus, it is possible that STING and TRIF may similarly cooperate in promoting RIPK1 activation and IFN- β synthesis. STING has also been proposed to respond to damaged DNA to induce IFN- β synthesis. This might serve as an indirect mechanism of STING participation in TRIF-RIPK1 kinase-dependent events. However, we believe this latter regulation is unlikely to be a major mechanism contributing RIPK1 kinase-dependent IFN- β synthesis because we observe considerable induction of IFN- β by LPS with zVAD in *Mikl*^{-/-} BMDMs, a system without measurable loss in cell viability (Fig. 3C, 3D). Nevertheless, further experiments are needed to further clarify the mechanism of STING involvement.

Protective host defenses mediated specifically by TRIF have been described in response to several Gram-negative bacterial pathogens (56–58). For instance, pulmonary infection by *Klebsiella* has been found to become significantly more deadly in TRIF-deficient mice (56). Conversely, preadministration of TLR3 agonist, poly(I:C), which engages TRIF-dependent signaling exclusively, has been shown to promote clearance of Gram-negative bacterial pathogens (58). Given these observations, our data raise the question whether previously described TRIF-dependent protective host responses are mediated by RIPK1/RIPK3 in vivo. Furthermore, our studies in macrophages suggest that TRIF/RIPK1/RIPK3-dependent responses may be either broadly or specifically targeted by bacterial virulence factors. Although this has been addressed in various viral infections (71–75), the regulation of RIPK1 and RIPK3 signaling by Gram-negative bacterial pathogen virulence factors is still poorly understood.

Accordingly, to our knowledge, our work is the first to demonstrate the capacity of RIPK1 and RIPK3 kinases to induce cytokine synthesis in response to Gram-negative bacteria. We report

FIGURE 8. RIPK1 and RIPK3 kinases engage the IFN-I pathway downstream of the adapter protein TRIF.



activation of RIPK1 and RIPK3 kinase-dependent IFN- β synthesis by *Klebsiella* and *Yersinia* mutants in vitro in the presence of zVAD. This observation is analogous to the regulation of RIPK1 previously noted in viral-induced RIG-I signaling in which caspase-8-dependent cleavage of RIPK1 attenuated IRF3 activation and IFN-I production (76). However, significantly, our data additionally reveal the propensity of avirulent bacterial strains to promote more efficient RIPK1 and RIPK3 kinase-dependent IFN- β induction compared with their WT pathogenic counterparts in vitro.

Data by Nogusa et al. (77) showed that RIPK3 was dispensable for the IFN-I transcriptional response induced by RIG-like receptor activation or influenza virus infection in mouse embryonic fibroblasts, in contrast to the central role of this protein in cell death induction during influenza infection in the same cell types (35). These data suggest that there may be important differences in the activation and consequence of the RIPK1 and RIPK3 kinase-dependent cytokine responses in cases of Gram-negative bacterial infections, which often occur extracellularly, versus intracellular viral infections, where cell death may be necessary to eliminate infected host cells and prevent viral replication. However, broader sets of pathogens and the roles of cell type-specific factors remain to be examined.

In this study, we describe a novel role for RIPK1 kinase in directing IFN- β synthesis in response to LPS in vivo. We find that RIPK1 kinase-dependent IFN- β synthesis may be elicited in analogous fashion in BMDMs using LPS with zVAD. Notably, we observed that RIPK1 kinase-dependent IFN- β synthesis in BMDMs also required RIPK3 kinase, but not MLKL, a requisite executioner of the necroptosis pathway. Instead, RIPK1 kinase-dependent IFN- β synthesis required the TLR4 adapter protein, TRIF, and downstream canonical IFN-I pathway intermediaries, including TBK1, IKK ϵ , and IRF3. It is important to note that RIPK1 and RIPK3 kinase-dependent IFN-I pathway activation in vitro was a delayed feature of TRIF-dependent signaling, suggesting that RIPK1 and RIPK3 kinases are not required for early endosomal translocation of TLR4 (78) (Fig. 8). Interestingly, we also noted that the intracellular nucleotide sensor, STING, likely participates in this regulation. Examination of detergent-insoluble cellular fractions or necrosome-like fractions in RIPK1

kinase-activating conditions suggested that the necrosome may represent a scaffold for the TRIF-IRF3 signaling axis, and/or that one or multiple IFN-I pathway intermediaries might be direct targets of RIPK1 and/or RIPK3. These data were further supported by coimmunoprecipitation studies. Finally, our observation that avirulent bacteria are capable of inducing RIPK1 and RIPK3 kinase-dependent IFN- β synthesis to a greater extent than their pathogenic counterparts in vitro prompts further examination of the role that RIPK1 and RIPK3 kinase-dependent IFN- β synthesis may play in the host response against bacterial infection.

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Disclosures

P.J.G. and J.B. are employees of GlaxoSmithKline. A.D. is a consultant for Denali Therapeutics. The other authors have no financial conflicts of interest.

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Supplemental Figures and Legends

Figure S1

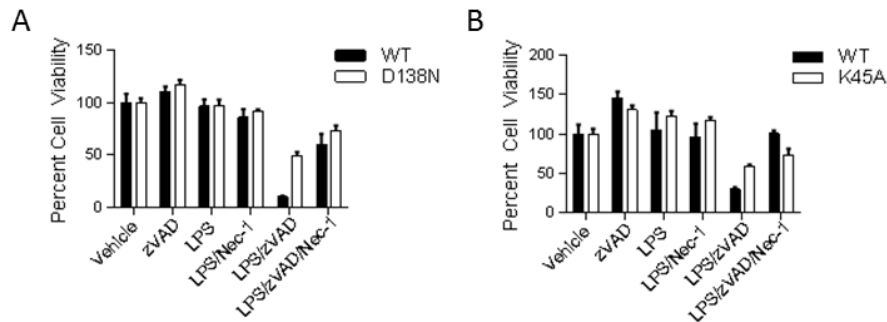


Figure S1: LPS with zVAD induces RIPK1 kinase-dependent cell death.

(A) Cell viability of wild type (WT) and D138N RIPK1 BMDMs (D138N) treated for 24 hours and evaluated by ATP assay. (B) Cell viability of WT and K45A RIPK1 BMDMs (K45A) treated for 24 hours and evaluated by ATP assay. Data are representative. Error bars reflect SD from the mean. BMDMs were treated with LPS=10 ng/mL, zVAD=50 μ M, and/or Nec-1s=30 μ M where indicated.

Figure S2

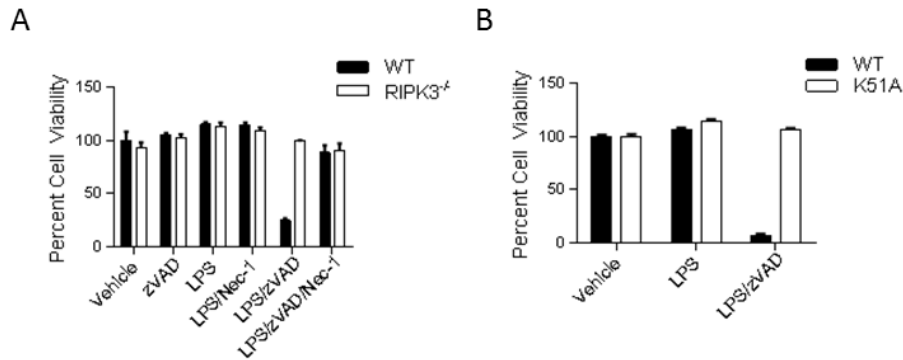


Figure S2: RIPK3 is required for cell death induced by LPS with zVAD.

(A) Cell viability of wild type (WT) and RIPK3 knockout (*Ripk3*^{-/-}) BMDMs treated for 24 hours and evaluated by ATP assay. (B) Cell viability of WT and K51A RIPK3 BMDMs (K51A) treated for 24 hours and evaluated by ATP assay. Data are representative. Error bars reflect SD from the mean. BMDMs were treated with LPS=10 ng/mL, zVAD=50 μ M, and/or Nec-1s=30 μ M where indicated.

Figure S3

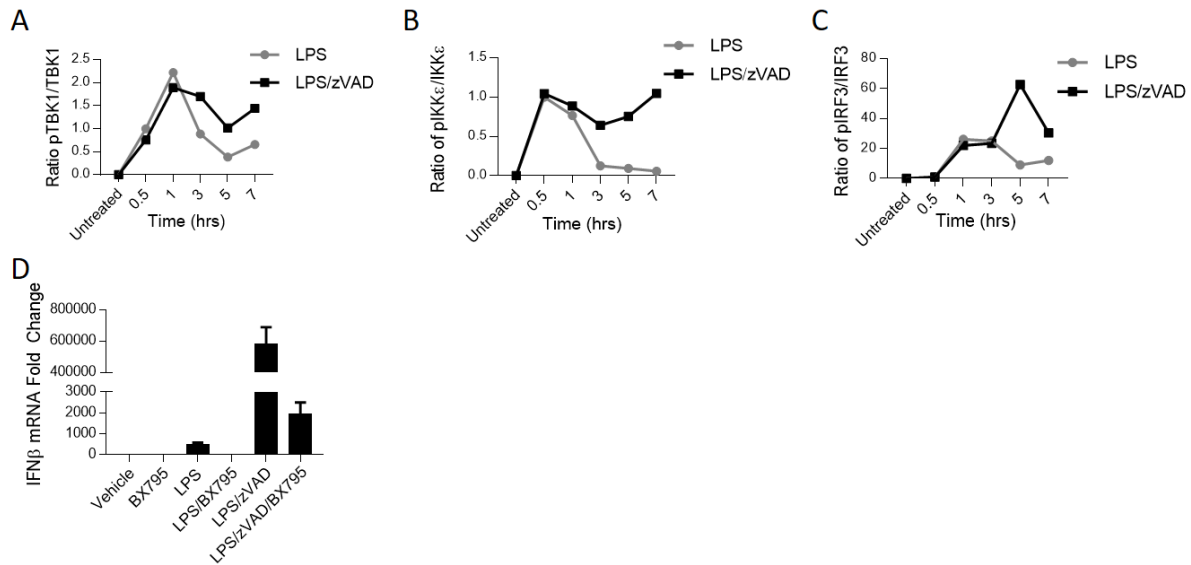


Figure S3: TRIF, TBK1/IKK ϵ , and IRF3 are important for LPS/zVAD induced IFN β synthesis.

(A-C) Densitometry analysis Western blot timecourse phosphorylation of TBK1 (A), IKK ϵ (B), and IRF3 (C) in wild type BMDMs. (D) qRT-PCR of *Ifnb* mRNA expression in wild type BMDMs treated with TBK1/IKK ϵ inhibitor (BX795) for 5-7 hrs. Data are representative. Error bars reflect SD from the mean. BMDMs were treated with LPS=10 ng/mL, zVAD=50 μ M, Nec-1s=30 μ M and/or BX795=1 μ M where indicated.

Figure S4

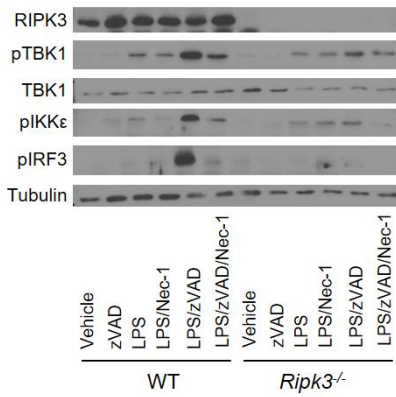


Figure S4: IFN-I pathway activation by LPS with zVAD is dependent on RIPK3.

Western analysis of TBK1, IKKε, and IRF3 phosphorylation in wild type (WT) and *Ripk3*^{-/-} BMDMs treated for 3-4 hours. Data is representative. BMDMs were treated with LPS=10 ng/mL, zVAD=50 μM, and/or Nec-1s=30μM where indicated.